

deceiving text detectors and leading to incorrect predictions. As corroborated by more detailed empirical studies, the proposed contrastive strategy greatly boosts the effectiveness and stability of the paraphrasing attack.

E Details about Machine Prompt

The prompts used in Figure 3 are as follows:

Machine Prompt 1

Repeat the following paragraph:

Machine Prompt 2

Rewrite the following paragraph in the tone of an AI assistant:

Machine Prompt 3

Paraphrase the following paragraph:

Machine Prompt 4

Rewrite the following paragraph:

F An Adaptive Defense Strategy

To alleviate the proposed threat, we implement an adaptive defense that adversarially trains OpenAI’s LLM-text classifier RoBERTa-large. We fine-tune the model for 10 epochs using 5k human texts and 5k machine texts (including both the original machine texts and those paraphrased by CoPA). We present the optimal performance at a proportion of 50% CoPA-paraphrased samples within the 5k machine texts in Table 8.

Table 8: Detection accuracy (at 5% FPR) of texts generated by GPT-3.5-turbo based on the XSum dataset.

Attack	w/o training	Adversarial training
No Attack	66.67	99.78
CoPA	22.67	78.00

The results indicate that the adaptive defense based on texts provided by our CoPA can alleviate the proposed threat to some extent.

G Human Prompts for Different Contexts

To better handle the different demands of contexts, we conduct a preliminary study by designing two alternative prompts as follows:

Human Prompt I for General Scenarios

Rewrite the following INPUT in human style, with varying sentence structures and replace common terms with nuanced synonyms. Maintain the meaning and avoid any repetition.

Human Prompt II for Academic Context

Rewrite the following INPUT in a human-written academic style, with varying sentence structures and replace common terms with nuanced synonyms. Maintain the meaning and avoid any repetition.

Here are the corresponding attack results:

Table 9: Attack results (at 5% FPR) of different human prompts using GPT-3.5-turbo generated texts on XSum.

Prompt	Fast-DetectGPT	TOCSIN	RoBERTa	R-Detect
Prompt I	24.67	43.33	15.33	18.67
Prompt II	24.33	39.67	5.33	14.33
CoPA	17.00	26.67	22.67	4.67

The quantitative results in Tab. 1 and demonstrations of paraphrased sentences show that CoPA is able to flexibly combine with prompts of various styles while maintaining high effectiveness. Users may design their own prompts based on our provided template to adapt to diverse writing styles.

H Evaluation on Text Quality

Apart from the attack effect, it is necessary to analyze the text quality of paraphrased sentences. Specifically, we first conduct a deeper linguistic analysis of the output texts in Table 10, including TTR (Type-Token Ratio) and MTLD (Measure of Textual Lexical Diversity) metrics for lexical diversity, Div_syn (Guo et al., 2024b) for syntactic variability. The results demonstrate the high quality of our generated sentences.

Table 10: Analysis of paraphrased sentences from different methods based on the XSum dataset.

Method	TTR↑	MTLD↑	Div_syn↑
No Attack	0.5734	111.4913	0.4700
Dipper	0.5511	74.8257	0.4648
CoPA	0.6746	145.8134	0.5593

We also provide a comprehensive evaluation of natural fluency and semantic consistency with addi-

Table 11: Comparison of our method with Dipper on text quality. The perplexity is calculated on GPT-neo.

Dataset	Text	Sim↑	Perplexity↓	GPT-4 Eval		Human Eval	
				Natural fluency↑	Consistency↑	Natural fluency↑	Consistency↑
XSum	Human	-	16.11	3.88	-	4.40	-
	Machine	-	8.857	4.33	-	4.94	-
	Dipper	86.67	14.76	3.74	3.76	4.25	4.26
	Ours	94	15.58	4.64	4.95	4.74	4.87
SQuAD	Human	-	19.52	3.60	-	3.98	-
	Machine	-	10.28	4.71	-	4.81	-
	Dipper	75.33	14.70	3.53	3.57	4.02	4.05
	Ours	88.67	17.77	4.56	4.91	4.56	4.79
LongQA	Human	-	27.54	3.48	-	3.75	-
	Machine	-	7.79	4.91	-	4.99	-
	Dipper	94.67	11.61	3.57	4.11	4.04	4.23
	Ours	95.33	13.08	4.57	4.99	4.53	4.81

tional key metrics, including text perplexity, GPT4-assisted evaluation, and human study. To conduct the human evaluation, we choose GPT-3.5-turbo as the source model and randomly select 100 pairs of texts from each dataset for human annotators. The evaluation criteria generally align with those in Dipper (Krishna et al., 2023), where we recruit 10 native English speakers from Amazon Mechanical Turk (MTurk) to perform the evaluation. We report the average scores to reduce subjective biases in Table 11. The results indicate that CoPA produces paraphrased texts with lower perplexity than authentic human-written texts, while achieving substantially better fluency and semantic consistency compared to those generated by Dipper, in both terms of GPT-4 assisted and human evaluation.

The instructions given to human annotators for *semantic consistency* align with those in Dipper, while we provide the scoring standard for *natural fluency* and the detailed prompt for GPT-4 auto evaluation as follows:

Instruction for Human evaluation:

Natural Fluency Scoring (1–5)

5. Excellent: Text flows perfectly naturally with varied, idiomatic phrasing. Grammar, word choice, and sentence structure appear completely native with zero awkwardness.

4. Good: Text reads smoothly with only minor and infrequent awkwardness. May contain 1-2 subtle non-native phrasings, but remains highly readable.

3. Fair: Generally understandable but contains noticeable unnatural phrasing. Some grammatical errors or awkward constructions occasionally disrupt flow.

2. Poor: Frequent unnatural phrasing and grammatical errors make reading difficult. Requires effort to understand in places.

1. Very Poor: Severely broken or unnatural English with major grammar issues. Often difficult or impossible to understand.

Prompt for GPT-4 evaluation:

(1) Task: Evaluate the natural fluency of a given sentence. Use a 5-point scale (5 = highest).

Natural Fluency (1-5):

1. Does the rewritten sentence flow naturally, avoiding awkward phrasing or redundancy?
2. Assess grammar, word choice, and readability (e.g., smooth transitions between clauses).
3. Penalize unnatural idioms or register mismatches (e.g., mixing formal and colloquial terms)

Output Format

Please provide the score for the metric.

Include a concise rationale (1-2 sentences per metric) highlighting specific strengths/weaknesses.

Example:

INPUT: "The deadline got pushed back because of unexpected tech issues."

OUTPUT: 4/5 (Colloquial tone matches intent; "pushed back" is natural but "tech issues" slightly informal).

(2) Task: Evaluate the semantic consistency of a rewritten sentence compared to its original version. Use a 5-point scale (5 = highest).

Semantic Consistency (1-5):

1. Does the rewritten sentence preserve the original meaning?
2. Check for critical information retention, logical coherence, and absence of distortion.
3. Deduct points for omissions, additions, or ambiguous interpretations

Output Format

Provide the score for the metric.

Include a concise rationale (1-2 sentences per metric) highlighting specific strengths/weaknesses.

Example:

INPUT:

Original: "The project deadline was extended due to unforeseen technical challenges."

Rewritten: "The deadline got pushed back because of unexpected tech issues."

OUTPUT:

5/5 (Key details retained; no loss of meaning).